



Full Length Article

Pediatric

A Randomized Clinical Trial Evaluating *Lactiplantibacillus Plantarum* for the Prevention of GI aGvHD: A Report From the Children's Oncology Group (ACCL1633)



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Gastrointestinal (GI) acute graft-versus-host disease (aGvHD) is a leading cause of non-relapse mortality following allogeneic hematopoietic cell transplant (alloHCT). Previous studies have suggested that the intestinal microbiome may influence the risk of GI aGvHD. We performed a Phase 3, randomized, placebo-controlled clinical trial to examine the effect of *L. plantarum* 299v (LBP 299v) in preventing GI aGvHD. Participants (N = 161 evaluable participants) received LBP 299v or placebo from the start of conditioning therapy to 56 days post alloHCT (D56). Blood, stool, and clinical data were collected until 120 days post-transplant (D120). The D120 cumulative incidences of stages 1–4 GI aGvHD were 16% and 15% ($P = .54$), and overall grades 2–4 aGvHD were 26% and 29%

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($P = .95$), *LBP 299v* and placebo groups, respectively. No patients developed *L. plantarum* bacteremia and no difference in serious adverse events was reported ($P = 1.00$). Administration of *LBP 299v* was associated with increased microbial diversity at D0 ($P = .02$) and reduced mucosal barrier injury at D7 ($P = .02$). Microbial signatures significantly differed between the groups; however, this was not associated with the investigated clinical outcomes. We conclude that administration of *LBP 299v* is safe among children and adolescents undergoing alloHCT but ineffective at preventing GI aGvHD.

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INTRODUCTION

Allogeneic hematopoietic stem cell transplant (alloHCT) is a curative treatment for children with select malignant and non-malignant conditions. Gastrointestinal (GI) acute graft-versus-host disease (aGvHD) is a significant cause of non-relapse mortality after alloHCT. Steroids are the primary treatment for GI aGvHD. While steroids can control aGvHD they have significant side effects and are not always effective. In steroid-refractory aGvHD, approximately 75% of patients will not respond to additional immunosuppressive therapy resulting in significant transplant-related mortality. There is a critical need to identify effective aGvHD prevention strategies that would translate into reduced needs for immune suppression and reduced transplant-related mortality [1].

Commensal organisms within the GI tract, collectively known as the intestinal microbiota, maintain physical, functional, and immunologic barriers within the GI tract. Conditioning regimens, antibiotic exposure, and prolonged use of total parenteral nutrition administered to patients undergoing HCT have the undesired consequences of damage to the GI epithelial layer and disrupt the balance of the microbial composition [2]. These interventions may lead to aberrant systemic immune responses [3] that promote pathogen colonization and mucosal invasion [4]. Observational cohort studies in adults undergoing HCT have identified microbial signatures and species associated with poorer survival and an increased risk of GI aGvHD [2,5,6]. Limited studies have investigated alterations in intestinal microbiota following pediatric allogeneic HCT. Masetti *et al.* found that children with low compared to high bacterial diversity prior to transplant experienced reduced survival (62.7% vs. 88.9%, respectively) as well as an increased incidence of Grades 2–4 aGvHD (44.4% vs. 20.0%, respectively). However, this association was not apparent based on diversity scores observed on the day of engraftment [7]. Gray *et al.* found that children with GI aGvHD had larger

fluctuations in the microbial composition compared to children without GI aGvHD [8] whereas a study performed in 88 children prior to and during HCT suggested that evaluation of the microbiome prior to HCT may be a more sensitive biomarker of adverse outcomes associated with HCT [9].

Replenishing the intestinal microbiota with commensals may potentially attenuate GI tract dysbiosis by preserving the intestinal microbiota or maintaining the integrity of the GI tract. In animal models, replenishing commensals with the genus *Lactiplantibacillus* prior to HCT substantially decreased the incidence and severity of aGvHD [10] and conferred a protective effect of aGvHD [11]. Our previous work found that the administration of the strain, *Lactiplantibacillus plantarum* 299v (herein, *LBP 299v*), was safe among children undergoing HCT [12] and may preserve the intestinal integrity of the GI tract, as indicated by serum citrulline [13]. While the animal models and pilot study data are encouraging, there remain no comparative studies to determine the efficacy of *LBP 299v* to prevent GI aGvHD in pediatric alloHCT recipients.

We performed a randomized, double-blind, placebo-controlled trial to determine the efficacy of orally administered *LBP 299v* in preventing GI aGvHD in children and adolescents undergoing alloHCT. We present results of analyses of the effect of *LBP 299v* on GI aGvHD, on additional select exploratory clinical outcomes, and biomarkers for the integrity of the GI tract [13–16]. We hypothesized that administration of probiotic therapy would prevent or minimize dysbiosis, preserving the microbiome and leading to a decrease in the incidence of GI aGvHD.

MATERIALS/SUBJECTS AND METHODS

Study Design

This was a multicenter, randomized, phase 3, double-blind placebo-controlled trial (ACCL1633) conducted by the Children's Oncology Group (COG) and approved by the National Cancer

Institute (NCI), Division of Cancer Prevention (NCT03057054). Study approval was obtained by the NCI's Central Institutional Review Board as well as the institutional review boards of all participating institutions located across the United States and Canada (N = 39). Informed consent and assent (when applicable) were obtained prior to study enrollment. The full protocol is available in [Supplementary Document 1](#). Recruitment took place from September 2018 to March 2022.

Participants

Eligible patients included were ≥ 2 and ≤ 25 years of age with a malignant or non-malignant condition that required treatment with alternative donor alloHCT. Alternative donors were defined as: (1) unrelated donor with 8/8 human leukocyte antigen (HLA) match or 1 or 2 HLA mismatch at HLA-A, HLA-B, HLA-C, and HLA-DRB1 loci or (2) related donor with 1 or more HLA mismatch, including haploidentical matching.

Exclusion criteria included: (1) receipt of probiotic therapy within 30 days prior to enrollment; (2) history of a severe gastrointestinal condition (e.g., Grade 4 neutropenic colitis or typhlitis, inflammatory bowel syndrome); (3) *Clostridioides difficile* (*C. difficile*) toxin positive diarrhea at enrollment; (4) an active infection at enrollment; (5) psychiatric concerns; or (6) allergy to third or fourth generation cephalosporins, carbapenems, and aminoglycosides given their activity against potential *LBP 299v* bacteremia.

Randomization and Blinding

Participants were randomized to receive daily *LBP 299v* or placebo using a 1:1 allocation with randomization stratified by stem cell source (cord blood vs. marrow or peripheral blood), HLA degree of match (complete vs. 1 vs. 2 or more mismatches), and GvHD prophylaxis (use of antithymocyte globulin/alemtuzumab vs. other). Enrollment and randomization occurred prior to the start of conditioning for transplant. Participants and clinicians were blinded to treatment assignment. *LBP 299v* and placebo had the same appearance, smell, and texture.

Administration, Adherence, and Colonization of *LBP 299v*

LBP 299v (mixture of *LBP 299v* and *LBP 299*) and placebo were provided by Probi® (Lund, Sweden; investigational new drug no. 17339). *LBP 299v* or placebo was prescribed for approximately 62–66 days beginning on the first day of conditioning therapy and continuing until 56 days post-HCT

(D56). The study dose, 1×10^8 colony forming units (CFU)/kg/day, of *LBP 299v* was based on our previous work describing the safety of *LBP 299v* among children undergoing HCT [12]. There were no dose adjustments for weight gain or loss during treatment. Purity and stability analyses were performed on *LBP 299v* and placebo annually to ensure no contamination and stable bacterial concentration. Supportive care was provided per institutional standards. Adherence was defined as the consumption of at least 80% of the prescribed dose. The methodology for assessing colonization of *LBP 299v* by the study arm may be found in [Supplementary Figures 1–2](#), and [Supplementary Table 1](#).

Clinical Outcomes

The primary outcome was stages 1–4 GI aGvHD within 120 days post-transplant, for which we sought to estimate and compare, across randomized arms, the proportional incidence by 120-days post-transplant. aGvHD was assessed utilizing the Modified Glucksberg Staging Criteria [17]. Confirmed, blinded central review of the case-report forms was used to validate aGvHD events ([Supplementary Table 2](#)). Planned exploratory outcomes included the proportional incidences of overall grades 2–4 aGvHD, diarrhea, diarrhea associated with a positive *C. difficile* toxin test, and cumulative days hospitalized for transplant, all by 120-day post-transplant and serum citrulline concentration across specimen collection timepoints. Post-hoc outcomes included time to neutrophil or platelet engraftment, primary graft failure, and 120-day aGvHD-free survival (GFS) (all patients) and aGvHD-and-relapse-free survival (GRFS) (patients with malignant diagnoses), which were separately analyzed using stages 1–4 GI aGvHD and grades 2–4 overall aGvHD. We additionally summarize treatment exposures, including days of antibacterial and antifungal agents, days of total parenteral nutrition (TPN), and days adherent to the study drug. Finally, Chao1 and Shannon alpha diversity indices were used as measures of microbiota diversity, and differential abundance analysis indicated variations within specific taxa [18–20]. For all exploratory outcomes except for the analysis of citrulline (only to Day 56 (D56) post-transplant), the follow-up period was identical to that of the primary aim.

Adverse Events (AEs)

AEs were reported through the study case report forms (CRFs) using the Common Terminology Criteria for Adverse Events (CTCAE) version 5.0. All serious AEs (SAE) reported through the CTEP Adverse Event Reporting System (CTEP-

AERS) were required for reporting if occurring within 30 days of administration of *LBP 299v* or placebo. SAE occurring after 30 days from the last administration of *LBP 299v*, or placebo were required for reporting if considered possibly, probably, or definitely related to *LBP 299v*. SAE included all AEs resulting in at least 24 hours of hospitalization or all CTCAE grade 3+ AEs. Any *L. plantarum*-positive blood cultures through Day 120 were required to be reported. CTCAE grade 2 bacteremia was not collected but grades 3–5 sepsis were collected.

Blood and Stool Collection and Analysis

Blood and stool specimens were collected at study entry (start of conditioning; SOC), day 0 (day of transplant), and days 7, 14, 28 (or discharge), 56, and 120 post-transplant. Specimens collected in the inpatient setting were transported to the institution's processing laboratory where stool was aliquoted and stored at -80°C and then batch-shipped on dry ice to Columbia University Irving Medical Center's microbiome laboratory. Specimens collected at home were stored in hazard bags with an ice pack and brought to the institution within 24 hours and processed in a similar manner as inpatient specimens.

Blood specimens were collected for measurement of serum citrulline. Citrulline measurement assays were performed utilizing previously described methodology [13]. Stool specimens were used to quantify the microbiome based on DNA extraction and metagenomic sequencing methods previously described [21]. Metagenomic sequencing data are publicly available through the NCBI Sequencing Read Archive (SRA) (PRJNA1157208). Alpha and beta diversity and differential abundance metrics were determined utilizing previously published methodology [19,20,22].

Statistical Analysis

All analyses were restricted to enrolled, eligible patients who received a stem cell infusion and did not withdraw before starting study treatment (herein "evaluable").

Primary Analysis

The primary analysis strategy is compliant with the protocol. We performed a 2-sample chi-square test to compare, across arms, the proportion of patients who experienced stages 1–4 GI aGvHD 120 days post-transplant. Patients who went off study prior to 120 days were counted as experiencing the primary outcome. Several protocol-specified sensitivity analyses of this approach

were implemented. We performed multivariable logistic regressions with adjustments for baseline patient demographic and clinical variables. We also implemented a Fine and Grey model of time-to stages 1–4 GI aGvHD, treating graft failure, relapse, and death as competing events, and censoring patients who withdrew consent for further data collection or reached 120 days post-transplant without experiencing an event, whichever occurred first.

Interim Analysis

The study was designed for 2 stages of analysis and was powered to detect a reduction in the proportion of stages 1–4 GI aGvHD within 120 days from 30% in patients randomized to placebo to 18% in patients randomized to receive *LBP 299v*. The first stage was conducted after 120 evaluable patients reached 120 days post-transplant. The study was to be referred to the data safety and monitoring committee for futility stopping if the p-value from a chi-square test comparing the incidence of stages 1–4 GI aGvHD across arms exceeded 0.3. If the futility rule was not met, the trial was to proceed to a second stage analysis to be conducted after accruing 454 total patients. The study ultimately stopped early due to futility after the first stage analysis. The study accrued beyond the 120 used for the first stage before closure, resulting in 161 evaluable patients.

Analysis of Baseline Information and Exploratory and Post-Hoc Outcomes

Descriptive statistics (count and proportion or mean and standard deviation (SD)) summarize patient clinical and demographic information. Exact tests and a t-test were used to compare corresponding distributions across arms. Analyses of planned exploratory outcomes: The same analytical methods were used for the primary outcome to analyze grades 2–4 overall aGvHD. Total days hospitalized were compared across arms using Wilcoxon rank sum tests. Binary outcomes of diarrhea and diarrhea associated with a positive *C. difficile* toxin test were compared using exact tests. Serum citrulline concentration were compared across arms and across study timepoints under a mixed model for repeated measures. Analyses of post-hoc exploratory outcomes: Time-to-engraftment, GFS, and GRFS were compared across arms using log-rank tests. We also report estimated Day 120 GFS and GRFS probabilities and corresponding 95% confidence intervals using Kaplan-Meier. The cumulative days of adherence to the study drug, days of TPN, and cumulative days of

antibacterial or antifungal agent exposures were analyzed using Wilcoxon rank sum tests. Chao1 and Shannon indices were compared across arms and timepoints under a mixed model for repeated measures.

Additional regression analyses of serum citrulline concentration, alpha diversity indices, and measures of *LBP 299v* colonization are described and summarized in the supplementary materials. Two-sided p-values are reported, and all analyses were performed using R software version 4.2.3.

RESULTS

Demographic and Clinical Characteristics

We enrolled and randomized 171 patients; 10 patients were not evaluable (*LBP 299v* arm 4; placebo arm 6, [Figure 1](#)). Of the 161 evaluable patients, no statistically significant differences for any demographic or clinical characteristics were identified between treatment groups ([Table 1](#)).

Adherence and Safety

Adherence to the intervention was high with $\geq 80\%$ of the prescribed doses administered in 70% and 79% in the *LBP 299v* and placebo arm, respectively ($P = .26$, [Supplementary Table 3](#)). The study

drug was also well tolerated. There were no significant differences in time to neutrophil and platelet engraftment, graft failure, relapse, and mortality between the study groups ([Table 2](#)). The number of adverse events and GI-specific adverse events were similar between both study arms ([Supplementary Tables 4–5](#)). Serious adverse events occurred in 7/80 (9%) and 8/81 (10%) of participants on the *LBP 299v* and placebo arm, respectively ($P=1.00$). There were no reported occurrences of *L. plantarum* bacteremia on either arm. Administration of antibiotics and antifungals was also similar between the 2 arms ([Supplementary Table 3](#)).

LBP 299v Stool Colonization and Microbial Diversity

DNA for sequencing was extracted from nearly all stool samples (*LBP 299v*, $n = 406$; placebo, $n = 446$ stool specimens). Four and 14 specimens from the *LBP 299v* and placebo groups, respectively, were not sequenced due to failed DNA extraction. On average, each sample contained 4,887,572 sequence counts and identified 9,151 bacterial taxa, with 343 of these taxa displaying a relative abundance of $\geq 0.0001\%$, as determined by Kraken2 using the standard database. Stool

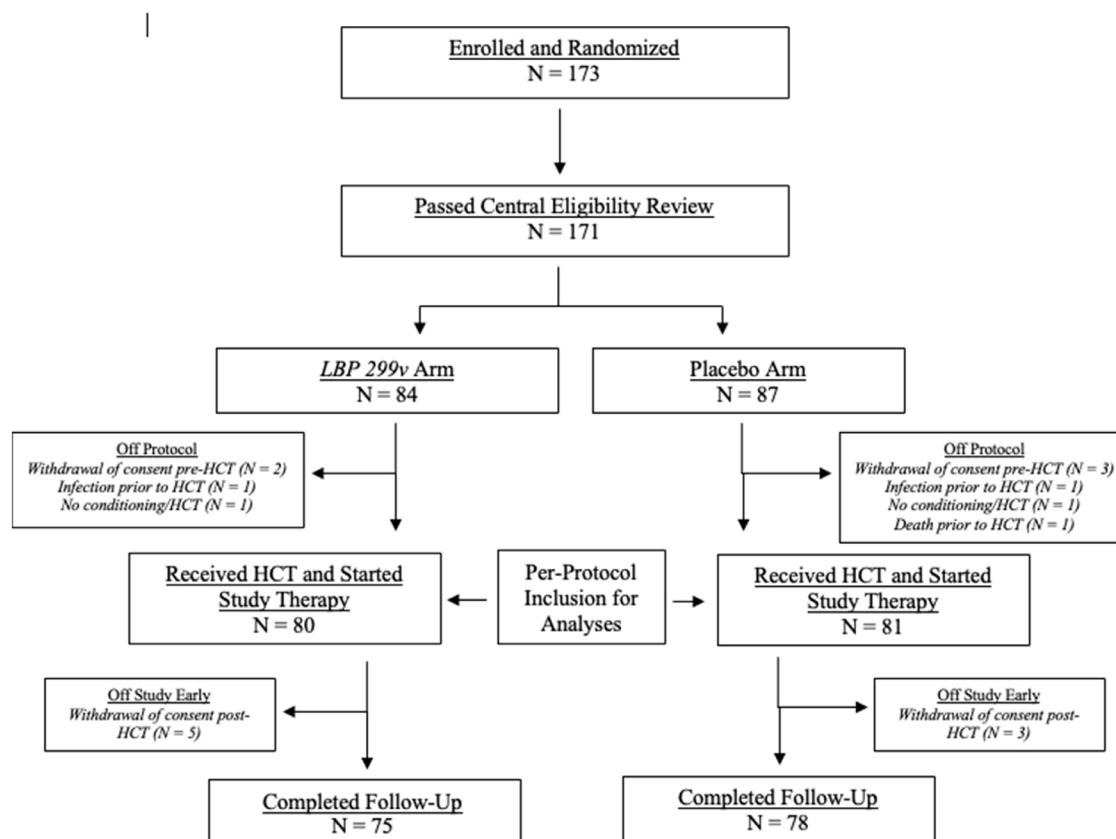


Figure 1. CONSORT diagram.

Table 1
Demographic and Clinical Characteristics of Study Participants

Variable	Values	All patients	LBP 299v	Placebo	P-value
Overall		161	80 (49.69%)	81 (50.31%)	
Sex – N (%)	Male	101 (62.7%)	53 (66.2%)	48 (59.3%)	.42
	Female	60 (37.3%)	27 (33.8%)	33 (40.7%)	
Age – Mean (SD)		12 (5.8)	11.3 (6)	12.7 (5.4)	.13
Race – N (%)	American Indian or Alaska Native	2 (1.2%)	2 (2.5%)	0 (0%)	.40
	Asian	8 (5.0%)	5 (6.3%)	3 (3.7%)	
	Black or African American	26 (16.1%)	13 (16.3%)	13 (16.0%)	
	Multiple reported	1 (1.0%)	0 (0%)	1 (1.2%)	
	Unknown	23 (14.3%)	14 (17.5%)	9 (11.1%)	
	White	101 (62.7%)	46 (57.5%)	55 (67.9%)	
Ethnicity – N (%)	Hispanic	32 (19.9%)	13 (16.2%)	19 (23.5%)	.48
	Non-Hispanic	118 (73.3%)	62 (77.5%)	56 (69.1%)	
	Unknown	11 (6.8%)	5 (6.2%)	6 (7.4%)	
Weight Category – N (%)	Under or normal weight	101 (62.7%)	49 (61.3%)	52 (64.2%)	.92
	Overweight	28 (17.4%)	14 (17.5%)	14 (17.3%)	
	Obese	32 (19.9%)	17 (21.2%)	15 (18.5%)	
Malignant diagnosis – N (%)	No	51 (31.7%)	29 (36.2%)	22 (27.2%)	.24
	Yes	110 (68.3%)	51 (63.7%)	59 (72.8%)	
Graft source – N (%)	Bone marrow	83 (51.6%)	42 (52.5%)	41 (50.6%)	.61
	Cord blood	7 (4.3%)	2 (2.5%)	5 (6.2%)	
	Peripheral blood stem cells	71 (44.1%)	36 (45.0%)	35 (43.2%)	
Donor sex – N (%)	Female	55 (34.2%)	27 (33.8%)	28 (34.6%)	1.00
	Male	78 (48.4%)	39 (48.8%)	39 (48.1%)	
	Unknown	28 (17.4%)	14 (17.5%)	14 (17.3%)	
TBI conditioning – N (%)	No	85 (52.8%)	46 (57.5%)	39 (48.1%)	.27
	Yes	76 (47.2%)	34 (42.5%)	42 (51.9%)	
Serotherapy conditioning – N (%)	No	77 (47.8%)	37 (46.2%)	40 (49.4%)	.75
	Yes	84 (52.2%)	43 (53.8%)	41 (50.6%)	
HLA degree of match – N (%)	Complete match	72 (44.7%)	36 (45.0%)	36 (44.4%)	.88
	1 Mismatch	32 (19.9%)	17 (21.3%)	15 (18.5%)	
	2 or More mismatches	57 (35.4%)	27 (33.8%)	30 (37.0%)	
Haploidentical donor – N (%)	Yes	52 (32.3%)	24 (30.0%)	28 (34.6%)	.61
GVHD prophylaxis – N (%)	Calcineurin Inhibitor	102 (63.4%)	56 (70.0%)	46 (56.8%)	.19
	Post-transplant cyclophosphamide	41 (25.5%)	18 (22.5%)	23 (28.4%)	
	T-cell depletion	12 (7.5%)	3 (3.8%)	9 (11.1%)	
	Other	6 (3.7%)	3 (3.8%)	3 (3.7%)	

colonization with LBP 299v on at least 1 evaluation was observed in 55 out of 78 (70.5%) and 3 of 77 participants (3.9%) in the LBP 299v and placebo arm, respectively, among those who submitted stool samples (Supplementary Table 3 and Supplementary Figures 2A–2B).

The average Chao1 and Shannon alpha diversity indices appeared different from the SOC to D14 post-transplant timepoints, with higher

average indices in patients randomized to LBP299v relative to placebo (Table 3 and Supplementary Figures 3–4). The difference was only statistically significantly different for the Shannon index at D0 (LBP 299v average: 2.6; Placebo average: 2.3; $P = .02$). Participants with a lower diversity at D0 experienced a higher degree of colonization compared to participants with a higher alpha diversity (i.e., higher alpha diversity

Table 2

Transplant and Clinical Related Outcomes Overall and Across Arms.

Variable	All patients	LBP 299v Arm	Placebo Arm	P-value
N Patients	161	80	81	-
Protocol Specified Outcomes				
GI aGvHD – N Events				
Stage 1	11	6	5	-
Stage 2	3	2	1	-
Stage 3	10	4	6	-
Stage 4	0	0	0	-
Stages 1-4*	24	12	12	.54
Stages 1-4 120-day cumulative incidence** estimate (95% CI)	0.16 (0.10,0.21)	0.16 (0.07,0.24)	0.15 (0.07,0.23)	-
Overall aGvHD – N Events				
Grades 1-4	66	29	37	-
Grades 2-4*	43	20	23	.95
Grades 2-4 120-day cumulative incidence** estimate (95% CI)	0.28 (0.21,0.35)	0.26 (0.17,0.36)	0.29 (0.20,0.40)	-
Post hoc transplant-related outcomes				
Time to engraftment (days) – median (IQR)				
Neutrophil	17 (14,21)	17 (14,19)	18 (14,23)	.78
Platelet	21 (17,30)	20 (16,30)	23 (18,32)	.52
Primary graft failure –N events (%)				
	6 (3.7%)	2 (2.5%)	4 (4.9%)	.64
GFS at 120-Days post transplant – probability (95% CI)				
For Stage 1-4 GI aGvHD	0.82 (0.76,0.88)	0.80 (0.72,0.90)	0.83 (0.76,0.92)	.60
For grades 2-4 overall aGvHD	0.70 (0.63,0.77)	0.70 (0.60,0.81)	0.69 (0.60,0.80)	.99
GRFS at 120-days post transplant*** – probability (95% CI)				
For Stage 1-4 GI aGvHD	0.78 (0.71,0.87)	0.75 (0.64,0.89)	0.81 (0.71,0.92)	.49
For grades 2-4 overall aGvHD	0.64 (0.55,0.74)	0.61 (0.49,0.76)	0.67 (0.56,0.80)	.45
Relapse – N Events				
	7	5	2	-
Overall mortality – N events				
	8	5	3	-

GFS: aGvHD-free-survival; GRFS: aGvHD-free relapse-free survival.

* aGvHD events are tabulated; p-value based on the primary analysis strategy.

** Cause-specific cumulative incidence accounting for competing events.

*** GRFS analysis restricted to patients with malignant diagnoses.

Table 3

Longitudinal Differences in Microbiome Alpha Diversity and Gut Integrity by Study Arm.

Timepoint	Chao1 index - mean (SD)			Shannon index - mean (SD)			Citrulline (uM) - mean (SD)		
	LBP 299v	Placebo	P-value	LBP 299v	Placebo	P-value	LBP 299v	Placebo	P-value
SOC	333.7 (11.4)	324.9 (37.9)	.11	2.8 (0.5)	2.8 (0.7)	.43	27.0 (13.2)	22.6 (9.6)	.03
Day 0	321 (46)	303.1 (56)	.06	2.6 (0.6)	2.3 (0.8)	.02	14.3 (9.3)	12.6 (9.1)	.36
Day 7	304.4 (44.3)	293.7 (52.8)	.15	2 (0.9)	2 (0.9)	.80	11.4 (7.1)	8.9 (5.7)	.02
Day 14	280.4 (61.3)	262.4 (76.8)	.22	1.7 (1)	1.5 (0.9)	.46	13.9 (10.1)	14.4 (8.4)	.67
Day 28	282.3 (61.2)	285 (59.7)	.86	1.9 (1)	1.9 (0.9)	.96	16.8 (9.6)	16.9 (9.4)	.90
Day 56	298.6 (73)	297.5 (71.1)	.89	2.3 (0.6)	2.2 (0.8)	.99	19.5 (9.6)	19.6 (11.5)	.48
Day 120	301.2 (88.7)	306.5 (72.5)	.60	2.7 (0.6)	2.4 (0.7)	.67	N/A	N/A	N/A

SOC, start of conditioning; D0, date of transplant; SD, standard deviation.

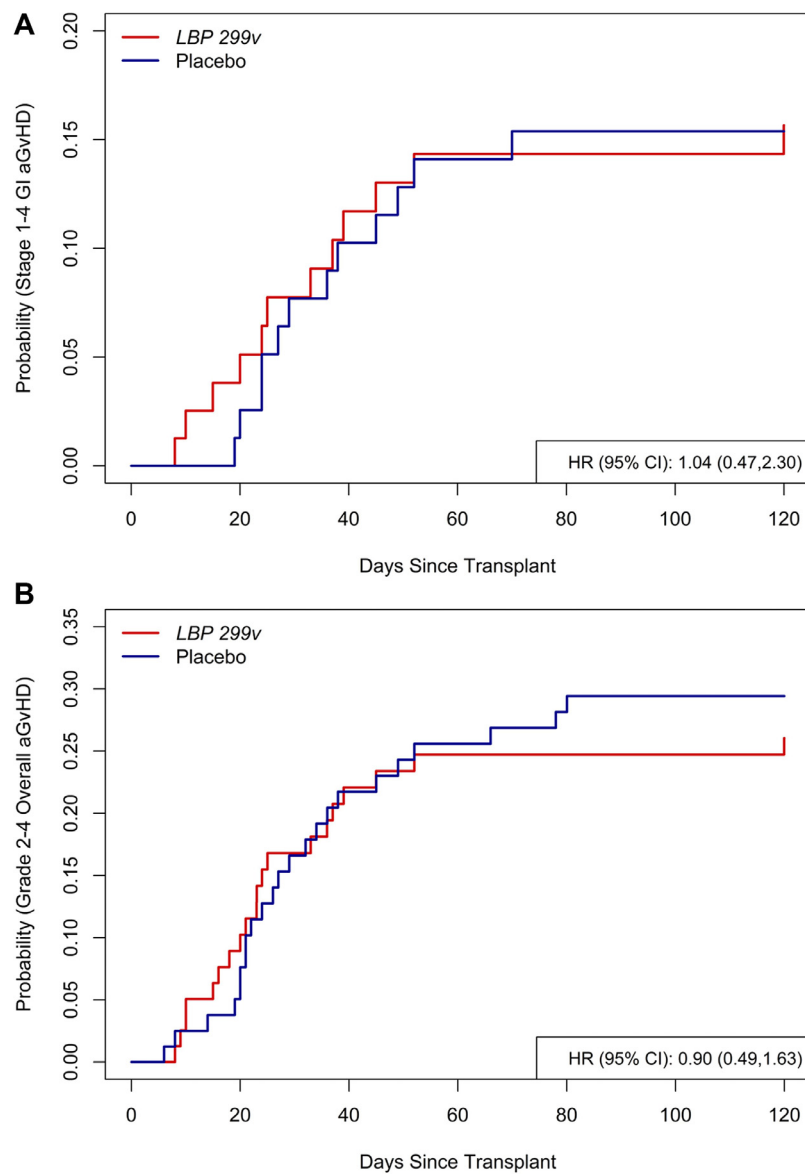


Figure 2. (A). Cumulative incidence at Day 120 of stages 1-4 GI aGVHD in the *LBP299v* and placebo arms. (B). Cumulative incidence at Day 120 overall grades 2-4 aGVHD in the *LBP299v* and placebo arms.

associated with decreased *LBP 299v* colonization), although the strengths of these associations were low (R^2 ranging from 0.19 to 0.33).

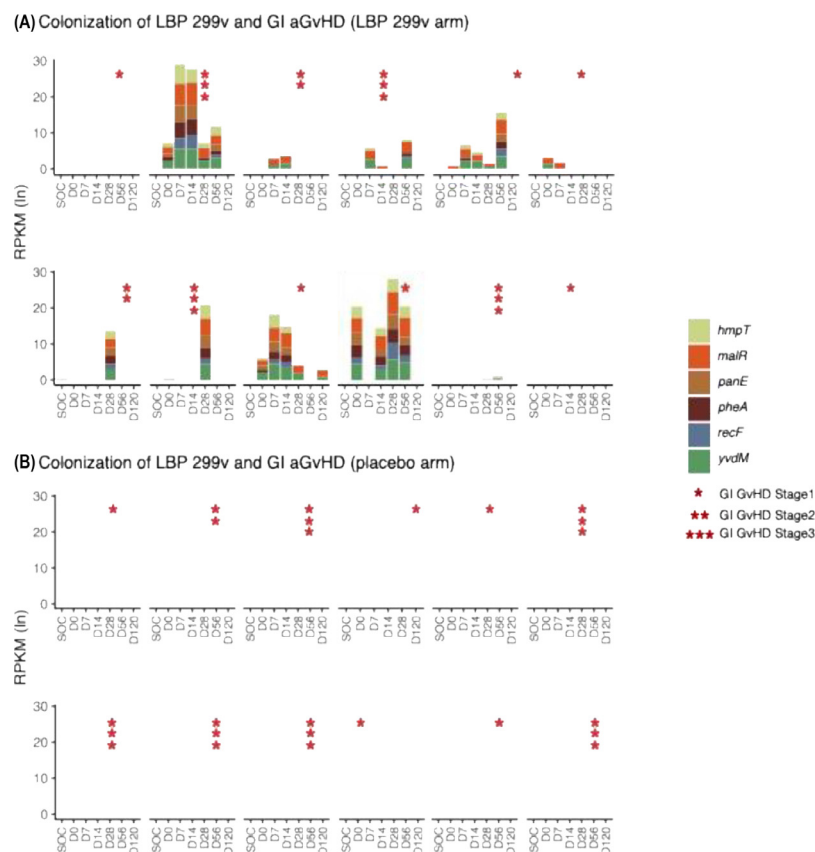
GI and Overall aGVHD

Using the protocol-specified primary analysis strategy, there were no significant differences between the *LBP 299v* and placebo arms in the proportion of patients experiencing stages 1-4 GI aGVHD by D120 (12 occurrences of GI aGVHD per arm, $P = .54$) or the proportion of patients experiencing Grades 2-4 overall aGVHD ($n=20$ and 23 , respectively, $P = .95$) (Table 2). Notably, there was no difference in Grade 3 GI aGVHD and no observed cases of Grade 4 aGVHD in either arm.

The absence of statistically significant differences remained after performing sensitivity analyses (time-to-event analyses: Figures 2A and B; adjusted regression analyses: Supplementary Table 6).

GFS and GRFS outcomes were similar across the 2 randomized arms (GI GFS: $P = .60$; Overall GFS: $P = .99$; GI-GRFS: $P = .49$; Overall-GRFS: $P = .45$) (Table 2 and Supplementary Figures 5a-5d). No differences were observed in the duration of TPN, days hospitalized, or incidence of *C. difficile*-associated diarrhea (Supplementary Table 3).

We summarized the occurrence of GI aGVHD among those who did and did not meet the criteria for *LBP 299v* colonization in the *LBP 299v* arm



Figures 3. (A and B). Colonization of *LBP 299v* by participants who developed GI GvHD. Each panel represents an individual patient, and the colors in the bar plots correspond to genes unique to *LBP 299v*.

(Figures 3A-B). Among the 55 *LBP 299v* arm patients who met the criteria for colonization, 9 (16%) experienced stages 1-4 GI aGvHD. Among the 23 patients who did not meet the criteria for colonization, 3 (13%) experienced stages 1-4 GI aGvHD. For the 4 participants on the *LBP 299v* arm who experienced stage 3 GI aGvHD, only 1 patient met colonization criteria at several timepoints before the onset of GI aGvHD. However, the collection of stool specimens was inconsistent, precluding comprehensive evaluation of colonization prior to GI aGvHD. We observed similar patterns of colonization in patients who developed GI aGvHD compared to patients who did not develop aGvHD (Supplementary Table 7).

Serum Citrulline

Higher mean serum citrulline concentrations were observed in the *LBP 299v* compared to the placebo arm at D7 ($P = .02$) when the nadir occurred (Table 3). A higher concentration of citrulline in the treatment arm at D7 remained after adjusting for citrulline values at SOC and body mass index ($P = .03$; Supplementary Table 8). Supplementary Figure 6 shows the subject-specific

and average serum citrulline values across study timepoints and indicates that average concentration was higher in the *LBP 299v* compared to placebo arm up to D14 post-transplant, after which point the curves converged.

We compared *LBP 299v* read counts (defined as the maximum log-RPKM achieved across timepoints) to the nadir of citrulline and alpha diversity indices at D14 (Supplementary Figures 7-9). There was no clear relationship between D7 serum citrulline and *LBP 299v* read counts. There was also no statistically significant relationship observed between either of the diversity indices or serum citrulline with GI aGvHD incidence (Supplementary Table 9).

DISCUSSION

We present the results of the first multicenter study assessing the efficacy and safety of probiotic *LBP 299v* administration to children and adolescents undergoing alternative donor (unrelated and/or HLA-mismatched) alloHCT. In this randomized Phase 3, double-blind placebo-controlled trial, we did not observe a significant reduction in GI aGvHD or overall grades 2-4 aGvHD in children

and adolescents receiving *LBP 299v* versus placebo. However, aligned with a previous study performed among adults undergoing HCT [23], our study did show that administration of *LBP 299v* probiotic was feasible and safe, as it did not result in any episodes of *L. plantarum* bacteremia or any increase in transplant-related toxicities. Furthermore, *LBP 299v* probiotic administration was associated with differences in measured microbial taxa compared to placebo and was associated with increased serum citrulline, suggesting potential for reduced intestinal microbial injury [13–16].

Previous literature, primarily in adults, has consistently reported an association between gut diversity or specific taxa and clinical outcomes in the setting of HCT [2,6,11,24,25]. Based on these observations, we sought to determine if modifying the microbiome through the administration of a probiotic would lead to improved clinical outcomes. One potential reason for a lack of clinical benefit is that a single species probiotic, *LBP 299v*, may not have been sufficient to overcome the insults that occur to the microbiome during HCT for a clinical benefit to be realized. Additionally, probiotic administration may have been suboptimal in dose, duration, and timing.

Despite excellent adherence to the protocol, stool colonization on at least 1 evaluation point was not observed in all patients on the *LBP 299v* arm (70.5%) nor at all timepoints. The duration of antibiotics administration was similar between the arms. It is plausible that if all participants had colonized in the *LBP 299v* arm at all timepoints, a beneficial effect may have been observed. Alternatively, colonization may be necessary prior to the beginning of the graft vs. host pathologic process, which likely starts as early as D0, for colonization to influence risk for GI aGvHD, such that much earlier administration of *LBP 299v* could be required [9]. However, such an analysis is difficult as this would require almost daily stool collection.

Our data suggest that low diversity of the microbiome may be associated with increased colonization of *LBP 299v*. It is possible children with lower diversity after conditioning (i.e., children with the greatest need for increased diversity) stand to obtain a greater benefit of probiotic intervention when compared to children with a higher diversity. Moreover, distinct signatures were observed among the *LBP 299v* arm compared to placebo. The mechanisms behind which supplementation elevates the presence of other strains and the interrelationship between these strains warrants additional research. Importantly,

several of these taxa have not been reported in previous studies in children undergoing HCT, thus their role on clinical outcomes warrants further inquiry.

We found that *LBP 299v* was associated with increased serum citrulline up to D7 post-transplant, suggesting a beneficial effect on the GI tract. However, the combination of the effects on the microbiome and the GI tract did not confer a clinical benefit for the primary or secondary outcomes in this Phase 3 trial. It is noteworthy that our study recruited children and adolescents 3–21 years of age representing significant variation in the development of an individual's microbiome. It is plausible that as the gut diversity and composition evolve during childhood and adolescence, different doses, strains, or combined interventions (i.e., pre- and pro-biotic interventions) may be warranted. Although we determined that it is safe to administer *LBP 299v* peri-HCT, it is possible that a beneficial effect on modulation of GI GVHD may only be accomplished via replenishment of 1 or more different microorganisms more important in the induction of alloreactivity.

Our results must be interpreted recognizing several strengths and limitations. This is the first multi-center clinical trial that administered probiotic therapy in a controlled fashion and systematically and longitudinally collected blood and stool from patients enrolled at several institutions located in the United States. Our study tested the probiotic annually to ensure the stability of colony-forming units and the purity of the product. Adherence to the study drug was similar to our previous study [12], and collection of blood and stool was exceedingly high. The primary reason for the lack of stool collection was the patient not passing stool, which minimizes bias in our results.

However, we acknowledge several limitations in the Phase 3 study. First, the dose of the *LBP 299v* was determined by a pilot study, where we conservatively established a dose that was considered to be safe. In this pilot study, colonization was defined as the presence of *L. plantarum* in the stool rather than confirmation of genes specific to *LBP 299v*, the approach used in the current study, which may explain conflicting results. Additional studies are needed to determine the safety and feasibility of administering higher doses, multi-strain probiotics, or a combination of pre- and probiotic therapy in children undergoing HCT. Our study was designed to recruit 454 patients; however, the study was stopped early after predetermined futility criteria were met. In turn, this resulted in only 24 GVHD events, limiting our

power to perform subgroup analyses, including stratification by antimicrobial therapy. Importantly, stool specimens were not obtained on all patients at all time points further limiting our ability to examine changes in microbial signatures in subgroups of enrolled patients.

In conclusion, this Phase 3 clinical trial demonstrated that *LBP 299v* can be safely administered to children and adolescents undergoing allogeneic HCT despite the lack of the prevention of GI aGVHD. We established significant variability in the colonization of *LBP 299v* and have identified several variables worthy of investigation to further understand the mechanisms behind probiotic therapy in the setting of alloHCT. However, given the low incidence of severe GI aGVHD in the control arm of this trial, additional analyses of baseline factors predisposing to GVHD are required in order to identify a suitable high-risk patient population for future prospective trials of microbiome modulation.

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SUPPLEMENTARY MATERIALS

Supplementary material associated with this article can be found in the online version at [doi:10.1016/j.jtct.2025.04.009](https://doi.org/10.1016/j.jtct.2025.04.009).

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